



United International University

Department of Computer Science and Engineering

CSE 2233: Theory of Computation Mid: Spring 2025

Total Marks: 30 Time: 1 hour and 30 minutes

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

Answer all the questions. The numbers on the right of the questions denote their marks.

1. Design DFAs that accept the following languages:

(a) $L = \{w \mid \text{every } w \text{ starts with even number of 'a' and ends with even number of 'b'}, \Sigma = \{a,b\}\}$ (3)

(b) $L = \{w \mid \text{every } w \text{ starts with 'aab' and contains 'bab' and ends with 'aa'}, \Sigma = \{a,b\}\}$ (3)

(c) $L = \{w \mid \text{every } w \text{ is made of only one type of alphabet}, \Sigma = \{a,b,c\}\}$ (3)

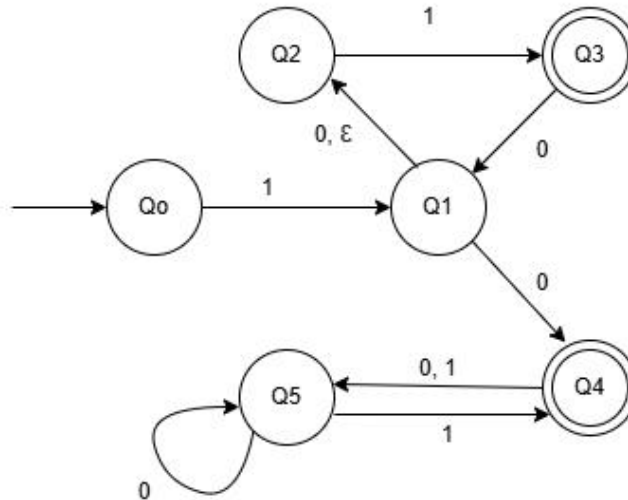
2. Design NFAs that accept the following languages:

(a) $L = \{w \mid w \text{ does not start with "ab", contains "bbc" and ends with "ac"}\}, \Sigma = \{a, b, c\}$ (3)

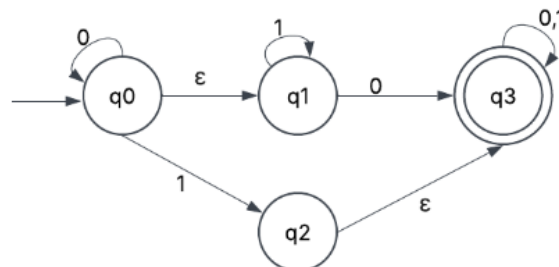
(b) $L = \{w \mid w \text{ starts with "yz" or "xz", contains "yzx" or "xyz" and ends with "zzy"}\}, \Sigma = \{x, y, z\}$ (3)

(c) $L = \{a^i b^j \mid \text{where } i \text{ is an odd number and } j \text{ is an even number}\}, \Sigma = \{a, b\}$ (3)

3. (a) Consider the following ϵ -NFA, and show with the help of NFA-tree whether the string "110011" is accepted or not. (3)



4. (a) Convert the following ϵ -NFA over alphabet $\Sigma = \{0,1\}$ to an equivalent DFA. Show both transition table and state diagram of the DFA. (6)



5. Design **Regular Expression** for the following languages:

(a) $L = \{w \mid w \text{ starts with abc, contains bac, and ends with bab over the alphabet } \{a, b, c\}\}$ (1)

(b) $L = \{w \mid w \text{ has even number of 0's over the alphabet } \{0, 1, 2\}\}$ (1)

(c) $L = \{w \mid w \text{ starts with ab and has even length or } w \text{ ends with cd and has odd length over the alphabet } \{a, b, c, d\}\}$ (1)